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School/Faculty:	**Razak Faculty of Technology and Informatics**		
Program name:	**Bachelor in Artificial Intelligence with Honours**		
Course code:	**SAIA 2143**	**Academic Session/Semester:**	**2025/2026 - 1**
Course name:	**Data Wrangling and Visualization**	**Pre/co requisite (course name and code, if applicable):**	**None**
Credit hours:	**3**		

Course synopsis	This course provides students with the foundational skills needed to clean, organize, and prepare raw data for analysis. Participants will gain hands-on experience using popular tools and techniques to tackle common challenges associated with messy datasets. The emphasis tools like spreadsheets,Python libraries and Power BI to make the learning experience accessible. This course will allow students acquire effective presentation skills to articulate and discuss data wrangling and visualization outcomes in both written and oral formats. The project allows students to apply their skills in a practical setting and showcase their understanding of data wrangling and visualization concepts.
Course coordinator (if applicable)	Dr Nilam Nur Binti Amir Sjarif

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
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Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (MQF Cluster)	Taxonomies Graduate Attributes	T&L methods	Assessment methods
CLO1	Explain the concepts and process including code and methodologies in data wrangling and visualization	PLO1 (C1)	C5 S	Lecture, Class Activity	Q T
CLO2	Construct data transformation using data wrangling and visualization techniques to clean and preprocess raw datasets , addressing missing values, outliers, and inconsistencies for analytics	PLO3 (C3A)	P4 IT	Lecture, Demonstration, active learning, Hands-on computer programming activity	Asg PR
CLO3	Demonstrate presentation skills effectively to articulate and discuss data wrangling and visualization outcomes	PLO5 (C3C)	A3 CC	Project Based Learning, Demonstration, active learning,	Pr

Taxonomies of Learning
Graduate Attributes: S - Scholar; CC - Constructive Citizen; IT - Innovative Talent
T – Test; Q – Quiz; Asg – Assignment; PR – Project; Pr – Presentation.

MQF Cluster Code
C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C= Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Enterpreneurial Skills, C5 = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active Learning	Conducted active learning through in-class activities such as buzz session, think pair share

2	Case Based Learning	Student work on real or hypothetical cases, applying their knowledge to analyze and solve problems. This method encouranges critical thingking and the application of concepts to practical situation
3	Project Based Learning	Students in a group of 3 - 4 are given a project to develop and implement the data wrangling and visualization by integrating theory and appropriate tools to selected focus group using pyhton programming and Power BI.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

Skill in design and developing a simple data wragling and visualization for day-to-day problem solving.
Skill in communication

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting/Topic	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	**Introduction to Data and Data Wrangling**	CLO1	2			1						2	5
Week 2	**Identifying and Handling Missing Data with Spreadsheets**	CLO1 CLO2					2			1		3	6
Week 3	**Dealing with Duplicates and Outliers**	CLO2	2			1						2	5
Week 4	**Introduction to Data Wrangling with Pandas (Python) Part 1**	CLO1	2			1					1	2	6
Week 5	**Introduction to Data Wrangling with Pandas (Python) Part 2**	CLO2	2			1						2	5
Week 6	**Advanced Data Wrangling Techniques (Part 1)**	CLO1	2			1						2	5
Week 7	**Introduction to Data Visualization**	CLO2	2			1						2	5
Week 8	SEMESTER BREAK												
Week 9	**Data Visualization with Matplotlib and Seaborn (Part 1)**	CLO2	2			1						2	5
Week 10	**Data Visualization with Matplotlib and Seaborn (Part 2)**	CLO2	2			1						3	6
Week 11	**Interactive Data Visualization with Plotly**	CLO1	2			1						2	5
Week 12	**Introduction to Dashboards (Part 1)**	CLO2					2			1		2	5
Week 13	**Introduction to Dashboards (Part 2)**	CLO2	2			1						3	6
Week 14	**Case studies on responsible in Data Visualization**	CLO2	2			1						2	5
Week 15	Project Presentation	CLO3				3						2	5
											SUB-TOTAL SLT :		**74**

Continuous Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
		Physical	Online (Synchronous)	Online (Asynchronous)	Others	

1	Quiz 1	15	1			6	7
2	Group Assignment	20			2	5	7
3	Test	25	2			8	10
4	Group Presentation	10	1			4	5
					SUB-TOTAL SLT :		**29**

Summative Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Individual Project	30			10	7	17
2							0
					SUB-TOTAL SLT :		**17**
					SLT for Assessment:		**75**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	33.33
B	% SLT for Online & Independent Learning Component :	66.67
C	%SLT for Online Component:	15.83
D	% SLT for All Practical Component:	0.00
D1	% SLT for F2F Physical Practical Component:	0.00
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)
Pytho Programming, Power BI, Spreadsheet

References (include required and further readings, and should be the most current)

Main references

a. Avinash Navlani, Armando Fandango, Ivan Idris (2021). Python Data Analysis: Perform data collection, data processing, wrangling, visualization, and model building using Python, 3rd Edition, Packt Publisher, ISBN: 978-1789955248
b. Sam Lau, Joseph Gonzalez, Deborah Nolan (2023).Learning Data Science: Data Wrangling, Exploration, Visualization, and Modeling with Python, O'Reilly Media Publisher, ISBN: 9781098113001

Additional references

Please refer to e-Learning
The Python Tutorial, https://docs.python.org/3/tutorial/
Python 3 Tutorial, https://www.w3schools.com/python/

Online: http://elearning.utm.my

Other additional information (if applicable)

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

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Signature:		**Signature:**	
Date:	09-Dec-23	**Date:**	09-Dec-23